Analysis of Algorithms: Lecture 1

MCQs with Answers (45) · Based on the lecture slides

★ = **most likely to come.** The slides say **Average case = Theta (Θ)**. Some books explain it differently, but give the answer from the slides in the paper.

Algorithm basics

1. An algorithm takes _____ and produces _____. ★

a) output, input
b) input, output
c) program, code
d) data, memory

Answer: b) input, output

2. How many ways are given in the slides to specify an algorithm? ★

a) 2
b) 3
c) 4
d) 5

Answer: b) 3 (Natural language, pseudocode, flowchart)

3. Pseudocode is a combination of: ★

a) flowchart and code
b) natural language and programming language constructs
c) machine language and English
d) diagrams and symbols

Answer: b) natural language and programming language constructs

4. A program is:

a) a flowchart
b) the expression of an algorithm in a programming language
c) a pseudocode
d) a data structure

Answer: b) the expression of an algorithm in a programming language

5. The science that studies and evaluates algorithms to select the best one is called:

a) Programming
b) Algorithmics
c) Data Science
d) Logic

Answer: b) Algorithmics

Features of algorithm

6. "An algorithm should end in a finite number of steps" is called: ★

a) Definiteness
b) Finiteness
c) Effectiveness
d) Output

Answer: b) Finiteness

7. "Each instruction should be clear" is called: ★

a) Finiteness
b) Definiteness
c) Effectiveness
d) Input

Answer: b) Definiteness

8. "Steps are sufficiently simple and basic" is called:

a) Definiteness
b) Finiteness
c) Effectiveness
d) Correctness

Answer: c) Effectiveness

9. Which is NOT a feature of an algorithm?

a) Finiteness
b) Definiteness
c) Input
d) Ambiguity

Answer: d) Ambiguity

Priori / Posteriori analysis

10. Theoretical estimation of resources required is called: ★

a) Posteriori analysis
b) Priori analysis
c) Experimental analysis
d) Average analysis

Answer: b) Priori analysis

11. Analysis done after implementing the algorithm on a target machine is: ★

a) Priori
b) Posteriori
c) Asymptotic
d) Theoretical

Answer: b) Posteriori

Complexity

12. Complexity of an algorithm is measured in terms of: ★

a) time only
b) space only
c) time and space
d) cost and lines of code

Answer: c) time and space

13. Space complexity = _____ ★

a) time + space
b) constant size + variable size
c) input + output
d) code size only

Answer: b) constant size + variable size

Types of algorithms

14. Randomized Quick Sort is an example of a _____ algorithm. ★

a) Heuristic
b) Approximate
c) Probabilistic
d) Greedy

Answer: c) Probabilistic

15. The algorithm based largely on optimism, where error cannot be controlled, is: ★

a) Probabilistic
b) Heuristic
c) Approximate
d) Divide and Conquer

Answer: b) Heuristic

16. The algorithm that specifies the error we are willing to accept is:

a) Heuristic
b) Probabilistic
c) Approximate
d) Dynamic

Answer: c) Approximate

Analysis and comparison

17. Which is NOT analyzed in an algorithm according to the slides?

a) Correctness
b) Amount of work done
c) Amount of space used
d) Colour of output

Answer: d) Colour of output

18. We usually focus on which running time? ★

a) Best case
b) Average case
c) Worst case
d) None

Answer: c) Worst case (Easier to analyze; crucial for games, finance, robotics)

19. In experimental studies, actual running time is measured using:

a) clock()
b) System.currentTimeMillis()
c) time.sleep()
d) printf()

Answer: b) System.currentTimeMillis()

20. Which is a limitation of experimental studies? ★

a) It is independent of hardware
b) The algorithm must be implemented
c) It considers all inputs
d) It uses pseudocode

Answer: b) The algorithm must be implemented

21. Theoretical analysis characterizes running time as a function of:

a) hardware speed
b) input size n
c) programming language
d) compiler

Answer: b) input size n

22. In the RAM model, each instruction takes: ★

a) variable time
b) unit time
c) zero time
d) n time

Answer: b) unit time

23. Comparing algorithms by execution time is not good because:

a) it is too slow
b) times are specific to a particular computer
c) it needs pseudocode
d) it uses Big-O

Answer: b) times are specific to a particular computer

Counting operations

24. arrayMax executes how many primitive operations in the worst case? ★

a) 2n
b) 4n
c) $6n - 1$
d) n^2

Answer: c) $6n - 1$

25. In for (i = 1; i < n; i++) the loop control executes:

a) n
b) 2n
c) $n - 1$
d) n^2

Answer: b) 2n ($1 + n + (n - 1) = 2n$)

26. Changing the hardware/software environment affects T(n) by:

a) changing the growth rate
b) a constant factor
c) an exponential factor
d) nothing at all

Answer: b) a constant factor (Growth rate does not change)

27. Two nested loops, each running N times, give:

a) O(N)
b) $O(N^2)$
c) O(log N)
d) O(1)

Answer: b) $O(N^2)$

Asymptotic notations

28. $10n^3 + 4n^2 - 4n + 5$ is: ★

a) O(n) b) $O(n^2)$
c) $O(n^3)$ d) $O(n^4)$

Answer: c) $O(n^3)$

29. Big-O represents: ★

a) lower bound b) upper bound
c) tight bound d) no bound

Answer: b) upper bound

30. Big-Omega (Ω) represents: ★

a) upper bound b) lower bound
c) average d) exact time

Answer: b) lower bound

31. f(n) = Θ(g(n)) if: ★

a) f(n) = O(g(n)) only b) f(n) = Ω(g(n)) only
c) f(n) = O(g(n)) and f(n) = Ω(g(n)) d) f(n) ≠ g(n)

Answer: c) f(n) = O(g(n)) and f(n) = Ω(g(n))

32. Worst case, best case and average case are represented respectively by (as in the slides): ★

a) Ω, O, Θ b) O, Ω, Θ
c) Θ, O, Ω d) O, Θ, Ω

Answer: b) O, Ω, Θ

33. The definition of Big-O is:

a) $f(n) \geq c \cdot g(n)$ b) $f(n) \leq c \cdot g(n)$
c) f(n) = g(n) d) f(n) > g(n)

Answer: b) $f(n) \leq c \cdot g(n)$

34. f(n) = O(g(n)) but f(n) ≠ Θ(g(n)) is:

a) Big-Omega b) Theta
c) little-o d) little-omega

Answer: c) little-o

35. f(n) = Ω(g(n)) but f(n) ≠ Θ(g(n)) is:

a) little-o b) little-omega (ω)
c) Big-O d) Theta

Answer: b) little-omega (ω)

36. (Past-paper style) $n^2 = O(n^3)$ is:

a) True b) False

Answer: a) True (n^3 is an upper bound of n^2)

Growth rate

37. $n^4 + 100n^2 + 10n + 50$ has the same rate of growth as: ★

a) $100n^2$ b) 10n
c) n^4 d) 50

Answer: c) n^4 (Elephant and goldfish example)

38. Which function grows the fastest? ★

a) N^3 b) N log N
c) 2^N d) N^2

Answer: c) 2^N (Exponential)

39. Which is the correct increasing order of growth?

a) $N < \log N < N^2$ b) $\log N < N < N \log N < N^2$
c) $N^2 < N < \log N$ d) $2^N < N^3 < N^2$

Answer: b) $\log N < N < N \log N < N^2$

40. N log N is called:

a) Linear b) Quadratic
c) N log N d) Cubic

Answer: c) N log N (N = Linear, N^2 = Quadratic, N^3 = Cubic, 2^N = Exponential)

41. Asymptotic analysis compares functions for:

a) small values of n b) large values of n
c) n = 0 d) negative n

Answer: b) large values of n

Miscellaneous

42. Which is NOT an example of input size?

a) size of an array b) vertices and edges in a graph
c) degree of a polynomial d) speed of CPU

Answer: d) speed of CPU

43. "A la russe" is a type of:

a) sorting algorithm b) multiplication algorithm
c) searching algorithm d) graph algorithm

Answer: b) multiplication algorithm (Halve one number, double the other)

44. 981 × 1234 using any classical multiplication algorithm gives:

a) 1210545 b) 1210554
c) 1201554 d) 1210454

Answer: b) 1210554

45. Which is NOT a classical multiplication algorithm in the slides?

a) English b) American
c) A la russe d) Binary Search

Answer: d) Binary Search

Short on time? Learn only the ★ MCQs

★ MCQ numbers: **1, 2, 3, 6, 7, 10, 11, 12, 13, 14, 15, 18, 20, 22, 24, 28, 29, 30, 31, 32, 37, 38**

5 common traps in MCQs

1. **Priori = theory**, **Posteriori = machine**. Do not reverse them.
2. **O = upper bound (≤)**, **Ω = lower bound (≥)**. Often reversed by mistake.
3. **Randomized Quick Sort = Probabilistic** (not Heuristic).
4. **Heuristic** = error cannot be controlled. **Approximate** = we decide the acceptable error.
5. **arrayMax = 6n − 1** operations.